



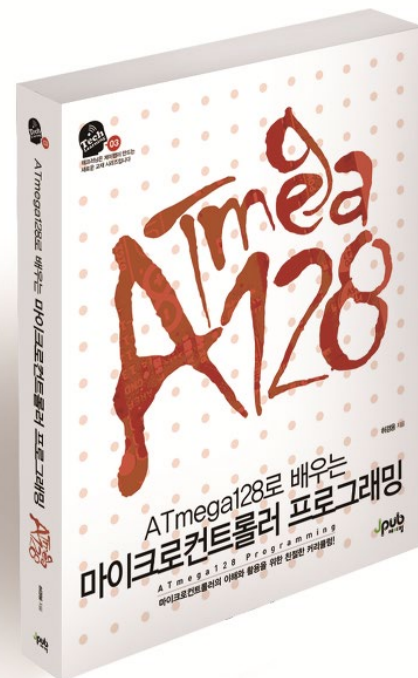
[01_Introduction]

한국폴리텍대학교 성남캠퍼스

1 강의 개요

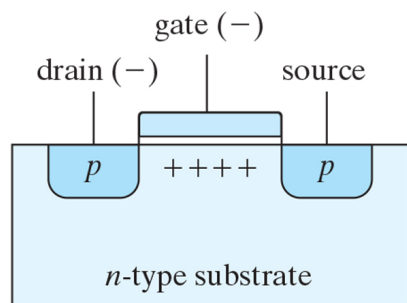
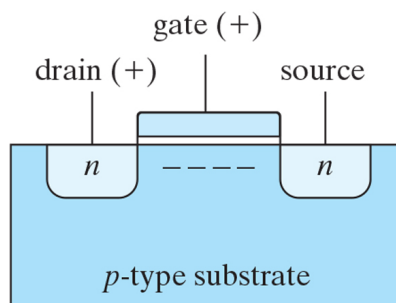
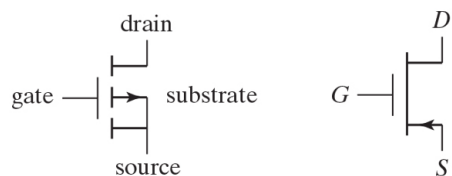
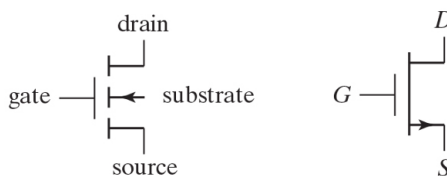
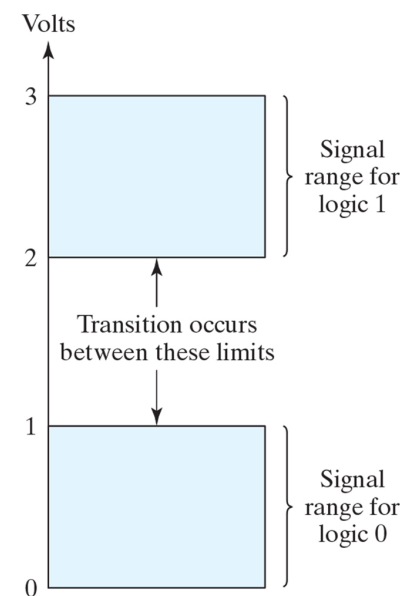
❖ 강의 참고자료

- 참고 교재는 다음과 같음
 - ATmega128로 배우는
마이크로컨트롤러 프로그래밍
 - 출판사 : 제이펍
 - 저 자 : 허경용
- 실습 관련 자료
 - PAMS - AER V9.1 아두이노 & MCU
실습 장비
 - 관련 PDF자료 제공



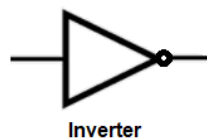
Basic unit

- The basic unit of a digital system is the bit, which is based on a switching device.
 - ✓ Switching device can control a signal flows and represent the state(0 or 1).

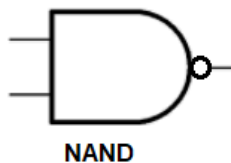
(a) *p*-channel(b) *n*-channel(a) *p*-channel(b) *n*-channel

Gate device

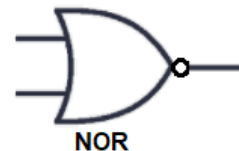
- The gate elements are defined by the combination of switching elements and wiring.



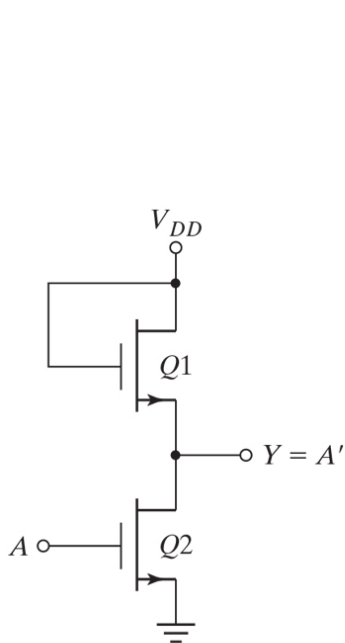
Inverter



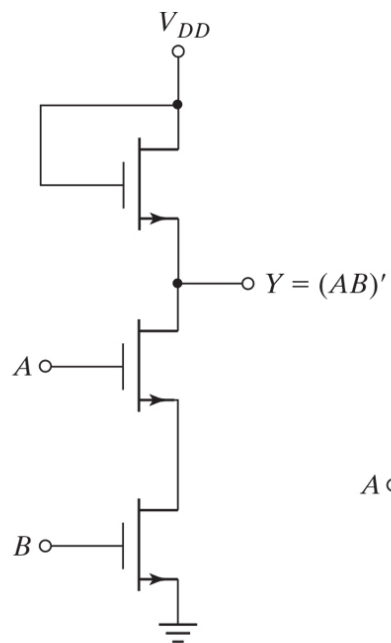
NAND



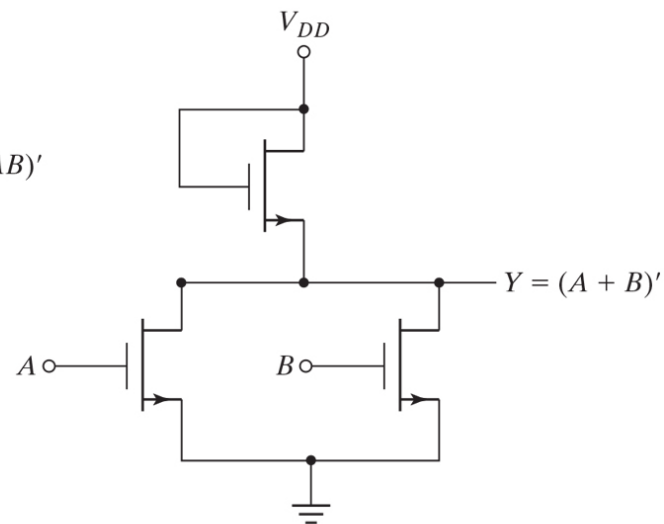
NOR



(a) Inverter



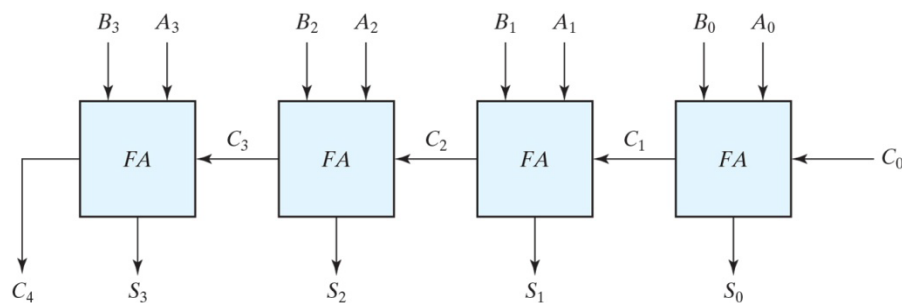
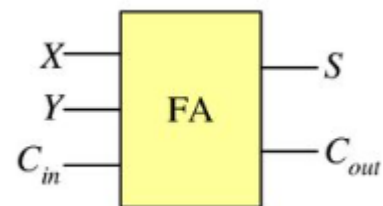
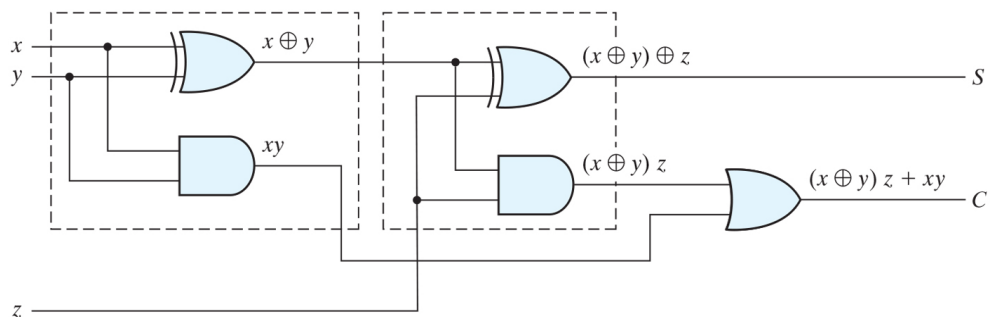
(b) NAND gate

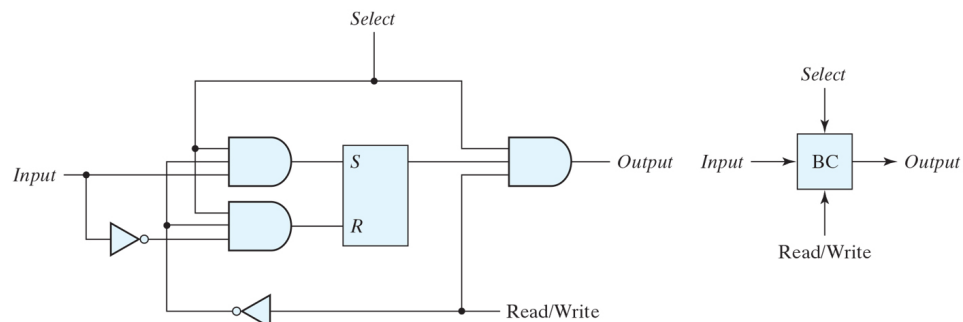
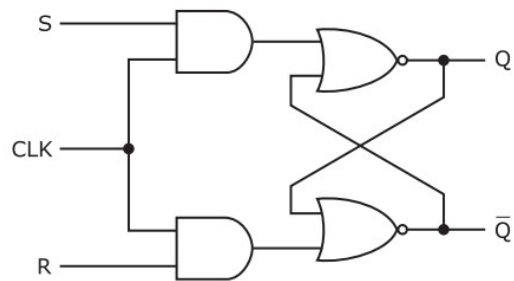


(c) NOR gate

Design Logic Circuit with Boolean Algebra

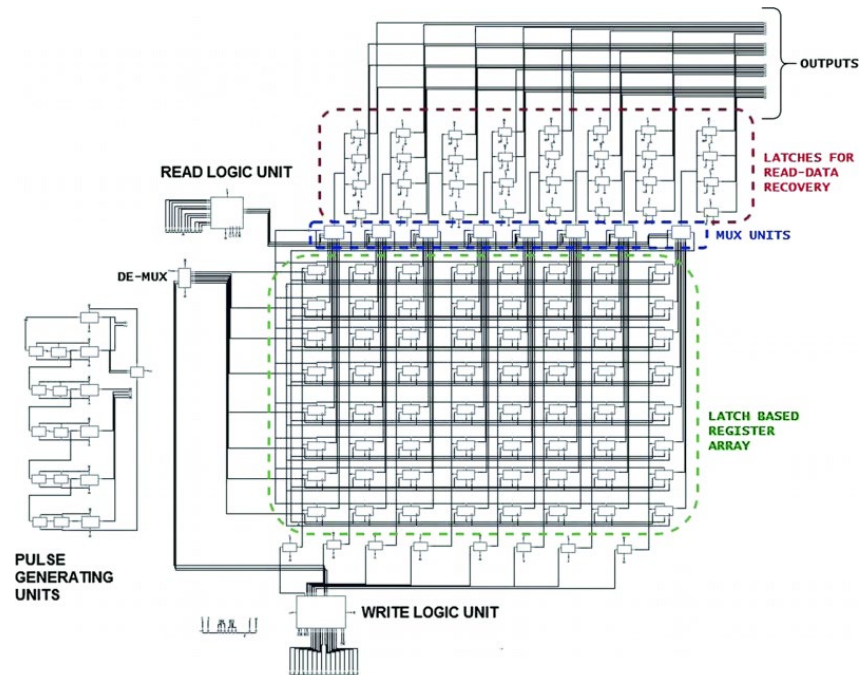
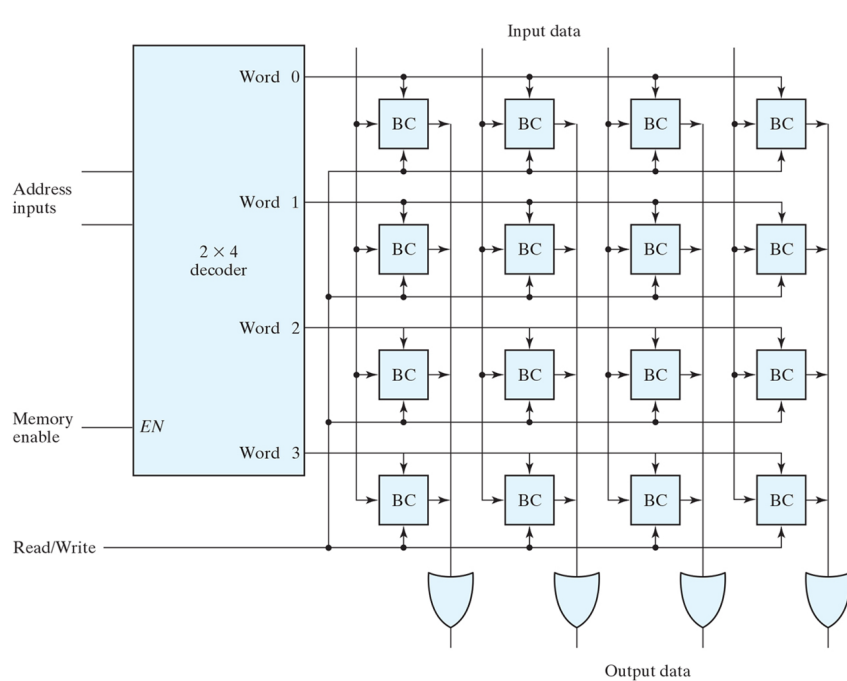
- It is possible to design a logic circuit that can perform a desired function through a combination of basic gate elements.
 - Combinational Logic Circuit (Adder, Mux, Encoder, Decoder etc.)
 - Sequential Logic Circuit (Register, Counter, Timer etc.)



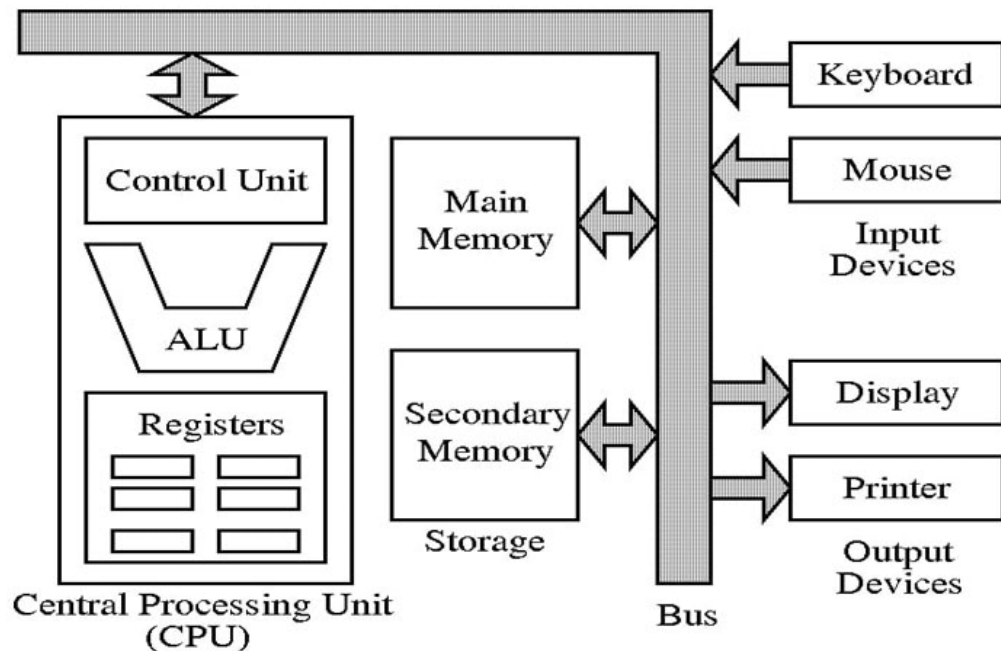
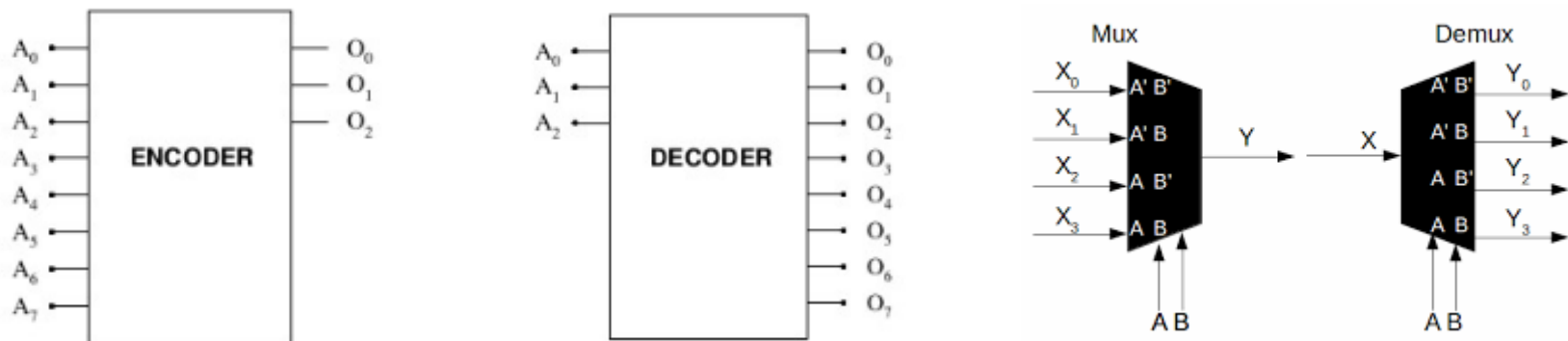


(a) Logic diagram

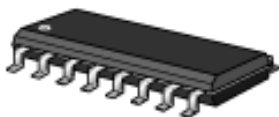
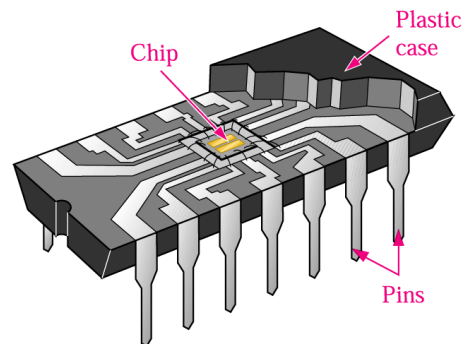
(b) Block diagram



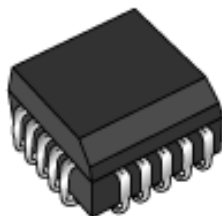
Digital IC(Integrated Circuit)



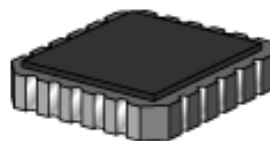
Digital IC(Integrated Circuit) Packages



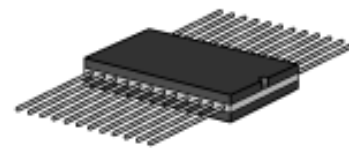
End view
(a) SOIC with
"gull-wing" leads



End view
(b) PLCC with
J-type leads



End view
(c) LCCC with no leads
(contacts are
part of case)



End view
(d) Flat pack (FP) with
straight leads

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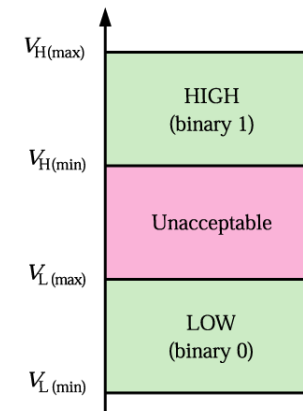
강의 개요

Number system

❖ Decimal number system

- We are used to the decimal number system which is of base 10. This means that it has ten digits and coefficients are multiplied by powers of 10

- Computers use the binary number system with base 2: it has two digits (0 and 1), and the coefficients are multiplied by powers of 2.



Base 10 Decimal	Base 2 Binary	Base 16 Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F
16	0001 0000	10
17	0001 0001	11
100	0110 0100	64
255	1111 1111	FF
1024	0100 0000 0000	400
65,535	1111 1111 1111 1111	FFFF

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강의 개요

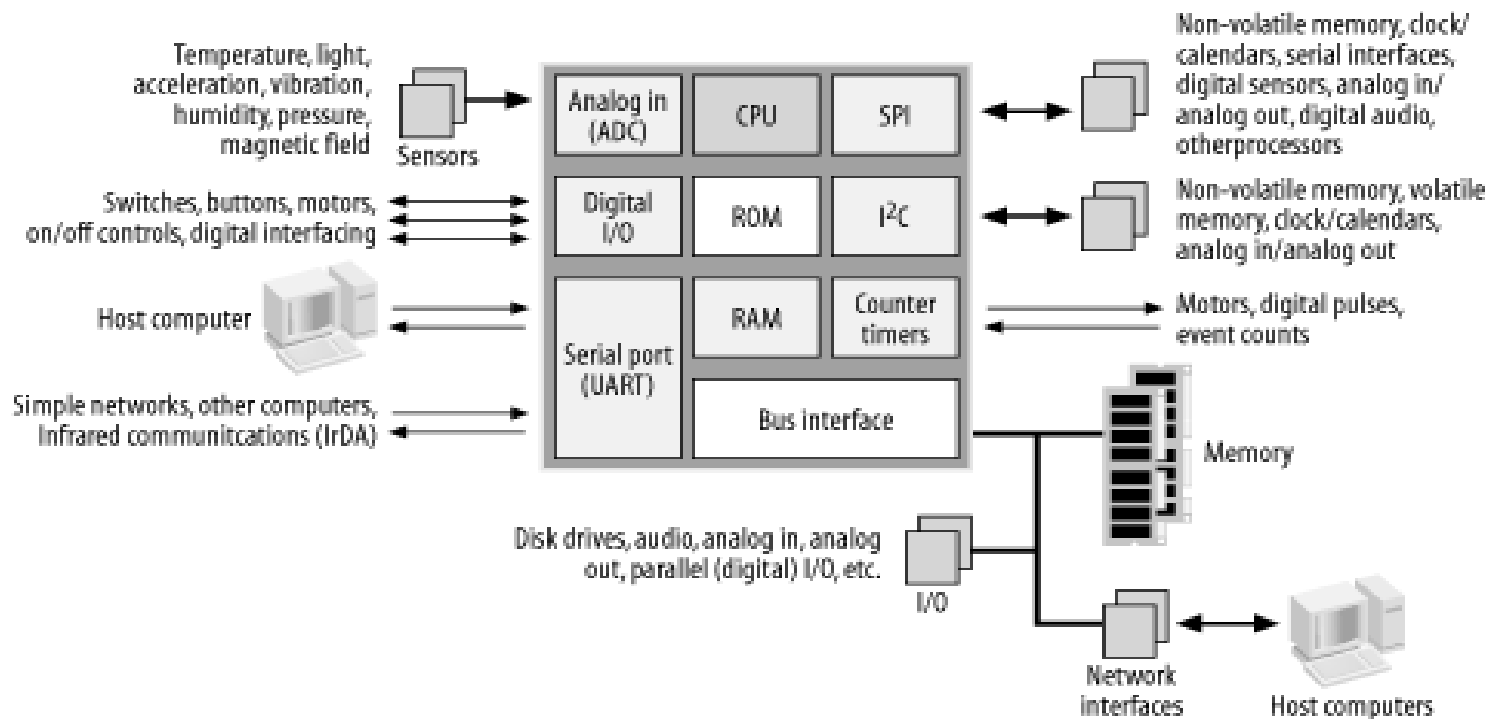
Digital Codes

- A code is a combination of bits of data to represent numbers, symbols, alphabetic characters, and other information.
- Code is a rule for representing data.
- These codes include various types such as BCD code, Gray code, and ASCII code according to the type.

Decimal Number	Binary				BCD				Excess-3				Gray			
	B ₃	B ₂	B ₁	B ₀	D	C	B	A	E ₃	E ₂	E ₁	E ₀	G ₃	G ₂	G ₁	G ₀
0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
1	0	0	0	1	0	0	0	1	0	1	0	0	0	0	0	1
2	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	1
3	0	0	1	1	0	0	1	1	0	1	1	0	0	0	1	0
4	0	1	0	0	0	1	0	0	0	1	1	1	0	1	1	0
5	0	1	0	1	0	1	0	1	1	0	0	0	0	1	1	1
6	0	1	1	0	0	1	1	0	1	0	0	1	0	1	0	1
7	0	1	1	1	0	1	1	1	1	0	1	0	0	1	0	0
8	1	0	0	0	1	0	0	0	1	0	1	1	1	1	0	0
9	1	0	0	1	1	0	0	1	1	1	0	0	1	1	0	1
10	1	0	1	0									1	1	1	1
11	1	0	1	1									1	1	1	0
12	1	1	0	0									1	0	1	0
13	1	1	0	1									1	0	1	1
14	1	1	1	0									1	0	0	1
15	1	1	1	1									1	0	0	0

Control Characters				Graphic Symbols											
Name	Dec	Binary	Hex	Symbol	Dec	Binary	Hex	Symbol	Dec	Binary	Hex	Symbol	Dec	Binary	Hex
NUL	0	0000000	00	space	32	0100000	20	@	64	1000000	40	'	96	1100000	60
SOH	1	0000001	01	!	33	0100001	21	A	65	1000001	41	a	97	1100001	61
STX	2	0000010	02	"	34	0100010	22	B	66	1000010	42	b	98	1100010	62
ETX	3	0000011	03	#	35	0100011	23	C	67	1000011	43	c	99	1100011	63
EOT	4	0000100	04	\$	36	0100100	24	D	68	1000100	44	d	100	1100100	64
ENQ	5	0000101	05	%	37	0100101	25	E	69	1000101	45	e	101	1100101	65
ACK	6	0000110	06	&	38	0100110	26	F	70	1000110	46	f	102	1100110	66
BEL	7	0000111	07	'	39	0100111	27	G	71	1000111	47	g	103	1100111	67
BS	8	0001000	08	(	40	0101000	28	H	72	1001000	48	h	104	1101000	68
HT	9	0001001	09	)	41	0101001	29	I	73	1001001	49	i	105	1101001	69
LF	10	0001010	0A	*	42	0101010	2A	J	74	1001010	4A	j	106	1101010	6A
VT	11	0001011	0B	+	43	0101011	2B	K	75	1001011	4B	k	107	1101011	6B
FF	12	0001100	0C	,	44	0101100	2C	L	76	1001100	4C	l	108	1101100	6C
CR	13	0001101	0D	-	45	0101101	2D	M	77	1001101	4D	m	109	1101101	6D
SO	14	0001110	0E	.	46	0101110	2E	N	78	1001110	4E	n	110	1101110	6E
SI	15	0001111	0F	/	47	0101111	2F	O	79	1001111	4F	o	111	1101111	6F
DLE	16	0010000	10	0	48	0110000	30	P	80	1010000	50	p	112	1110000	70
DC1	17	0010001	11	1	49	0110001	31	Q	81	1010001	51	q	113	1110001	71
DC2	18	0010010	12	2	50	0110010	32	R	82	1010010	52	r	114	1110010	72
DC3	19	0010011	13	3	51	0110011	33	S	83	1010011	53	s	115	1110011	73
DC4	20	0010100	14	4	52	0110100	34	T	84	1010100	54	t	116	1110100	74
NAK	21	0010101	15	5	53	0110101	35	U	85	1010101	55	u	117	1110101	75
SYN	22	0010110	16	6	54	0110110	36	V	86	1010110	56	v	118	1110110	76
ETB	23	0010111	17	7	55	0110111	37	W	87	1010111	57	w	119	1110111	77
CAN	24	0011000	18	8	56	0111000	38	X	88	1011000	58	x	120	1111000	78
EM	25	0011001	19	9	57	0111001	39	Y	89	1011001	59	y	121	1111001	79
SUB	26	0011010	1A	:	58	0111010	3A	Z	90	1011010	5A	z	122	1111010	7A
ESC	27	0011011	1B	;	59	0111011	3B	[	91	1011011	5B	{	123	1111011	7B
FS	28	0011100	1C	<	60	0111100	3C	\	92	1011100	5C		124	1111100	7C
GS	29	0011101	1D	=	61	0111101	3D	]	93	1011101	5D	}	125	1111101	7D
RS	30	0011110	1E	>	62	0111110	3E	^	94	1011110	5E	~	126	1111110	7E
US	31	0011111	1F	?	63	0111111	3F	_	95	1011111	5F	Del	127	1111111	7F

Block Diagram of an embedded computer



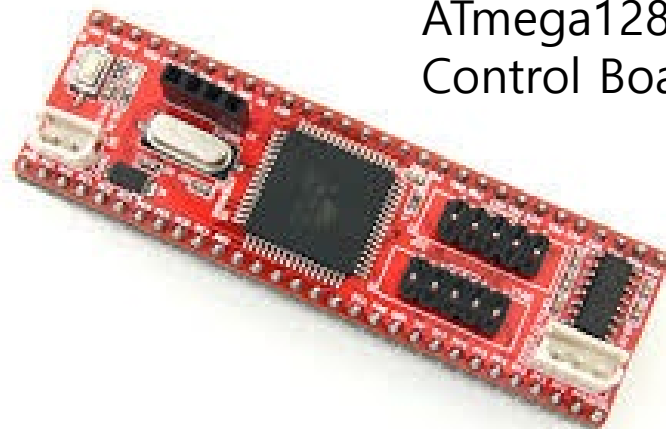
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강의 개요

■ 마이크로 컨트롤러



ATmega128
(Micro-Controller)



ATmega128
Control Board



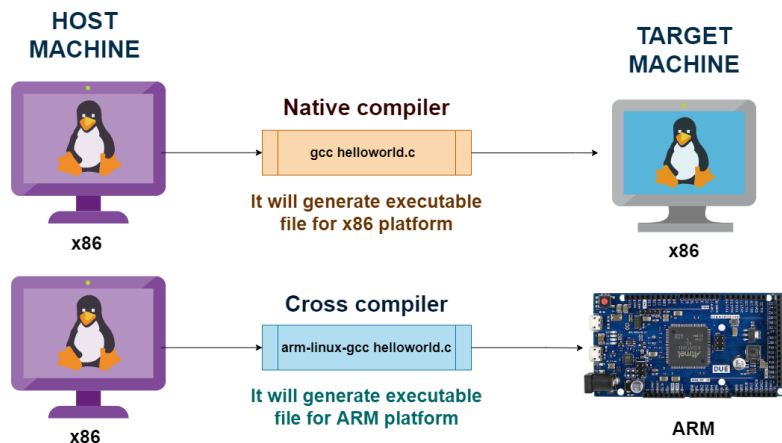
ATmega128
Training Module

■ 마이크로 컨트롤러

- 다른 시스템의 일부로 사용되는 경우가 흔함
- $uC = uP + \text{메모리} + \text{입출력 인터페이스}$
- 컴퓨터의 메인보드와 하드디스크 기능을 하나의 IC 칩으로 집적시켜 만든 반도체 소자(단일 칩 마이크로컴퓨터, 마이컴)
- 작고 간단한(마이크로) 제어장치(컨트롤러)를 만들기 위한 목적으로 특화된 마이크로프로세서의 한 종류

Development Platform

- Personal computers are used as development platforms for microcontroller projects.
- Development computers, usually personal or workstation computers, use a microprocessor as their principal computing engine.
- A development platform is required to run embedded system development software such as assemblers, compilers, editors and simulators which require the processing power and memory capabilities of a desktop personal computer or workstation.



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강의 개요

■ 마이크로 컨트롤러



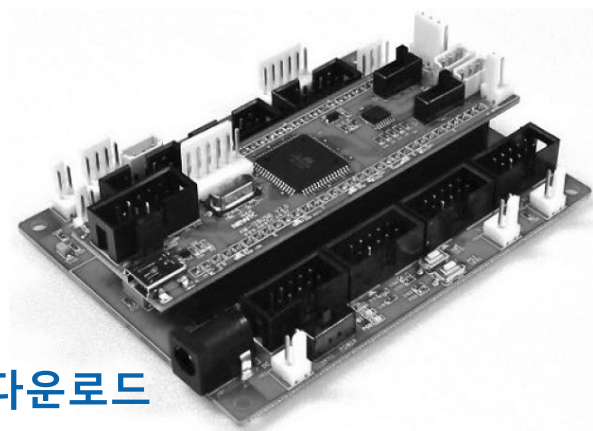
교차 컴파일러가 설치된
개발 시스템

*.hex

개발된 기계어 파일을
다운로드



ISP 또는 Serial 방식으로 다운로드



개발된 프로그램을 실행할
목적 시스템

- Cross Compiler : Atmel Studio

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강의 개요

■ 확인 사항

- C Programming 및 전자회로 배경지식 확인
- Test 프로그램 작성 및 학습현황 파악

Q & A
